

Each criterion was evaluated against every other using a pairwise comparison method. Values were assigned in 0.25 increments between 0 and 1, where:

- A value of 0.5 indicates equal importance.
- Values of 0.25 or 0.75 indicate one criterion is moderately more important than the other.
- Values of 0 or 1 reflect extreme preference toward one criterion, rendering the other negligible.

The comparisons are symmetric such that the sum of each pair equals 1. Total weights were summed and normalized to create priority values for each criterion.

Criteria	Mass	Power Demand	Complexity/ Cost	Accuracy	Environmental	Weight	Normalized
Mass		0.25	0.5	0.25	0.25	1.25	0.12
Power Demand	0.75		0.75	0.25	0.5	2.25	0.23
Complexity/ Cost	0.5	0.25		0.25	0.25	1.25	0.12
Accuracy	0.75	0.75	0.75		0.75	3	0.3
Environmental tolerance	0.75	0.5	0.75	0.25		2.25	0.23
Total						10	

	Mass	Power Demand	Complexity/ Cost	Accuracy	Environmental
5	< 0.15 kg total	< 1 W	< \$40k, COTS, no new qualification	Heading knowledge < 0.15 deg	Space-hardened, no moving parts, > 100 krad and 90 K proven
4	0.15 - 0.4 kg	1 - 2 W	\$40 - 100k, minor	0.15 - 0.35 deg	Minor degradation at
3	0.4 - 0.8 kg	2 - 5 W	\$100 - 250k, custom	0.35 - 0.7 deg	Needs derating or added
2	0.8 - 1.5 kg	5 - 10 W	\$250k+, new	0.7 - 1.5 deg	Noticeable degradation,
1	> 1.5 kg	> 10 W	One-off, requires a	> 1.5 deg	Strongly affected or

Decision Criteria		Analysis of the absolute heading reference for surface navigation					
		Fine sun sensor pair + onboard ephemeris		COTS small star tracker		Navigation-grade FOG gyrocompassing (no optical fix)	
Criteria	Weight	Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
Mass	0.12	5	0.6	4	0.48	2	0.24
Power Demand	0.23	5	1.15	3	0.69	2	0.46
Complexity/ Cost	0.12	4	0.48	2	0.24	1	0.12
Accuracy	0.3	4	1.2	5	1.5	3	0.9
Environmental tolerance	0.23	5	1.15	2	0.46	2	0.46
Total Score		4.58		3.37		2.18	

The fine sun sensor pair scores 4.58 against 3.37 for the star tracker and 2.18 for gyrocompassing with a navigation-grade FOG. Note that this trades the absolute heading reference only. The MEMS IMU stays in the design either way, because something has to propagate attitude between fixes and damp rate during a turn. The star tracker wins on raw accuracy and loses everywhere else: it is the one option whose detector degrades measurably under the Jovian dose, it needs a baffle against Jupiter, which subtends about 11.8 degrees from the surface and never sets on the sub-Jovian hemisphere, and it costs more than the rest of the GNC sensor suite combined. Gyrocompassing is the interesting failure. A 0.01 deg/h FOG can in principle find north off Europa's 4.22 deg/h rotation, but that signal is 28% of what the same instrument would see on Earth, and the fiber coil and light source are not qualified at 90 K or at 100 krad. It also draws 5 to 10 W continuously against a budget that already sits at 2006.5 Wh/sol. The sun sensor is the cheap answer: at 5.2 AU the Sun delivers only 50 W/m2, but its angular diameter collapses to 0.103 deg, so a photodiode sensor is looking at a sharper point source than it would at Earth. Two of them cost 0.12 kg and under 1 W, and Cassini flew the same class of sensor further out than Jupiter. At +/-0.1 deg on the Sun vector the heading solution lands at +/-0.25 deg once ephemeris and mounting alignment are folded in, which is what REQ.11.01 needs and what keeps the 50 m localization budget closing at 0.46 m against the 0.5 m limit in REQ. 02.01. Selected: fine sun sensor pair with onboard ephemeris.